PlugPlan SG — Refined Dialog Map (Design Model)

NTU SC2006 Software Engineering | Lab 3 Deliverable | Team **SCE3-04**

Reference Standards: Christopher Fox (2006), *Introduction to Software Engineering Design*, Chapter 13.3 'Dialog Maps', p. 420; Bernd Bruegge & Allen H. Dutoit, *Object-Oriented Software Engineering*, Chapter 7.4.
Architectural Alignment: Layered Client-Server Architecture (Presentation, Application, Domain, Integration, Persistence); TypeScript / Next.js / Supabase PostgreSQL Skeleton.

1. Executive Summary & Analysis-to-Design Evolution

In **Lab 2 (Requirements Analysis)**, an initial Dialog Map established an analysis-level UI State Machine defining abstract boundary interactions between the system's primary actors (EV Driver and Moderator) and high-level frames (S0–S6). In **Lab 3 (Design and Implementation)**, this analysis model is systematically transformed into a concrete **Design Model** adhering to Christopher Fox's formal state-event-guard-action dialog acceptor semantics, direct React presentation class bindings (PlugPlanApp, AuthScreen, JourneyMap, PlacePicker, Modal), granular modal lifecycles, and resilient distributed error handling.

Aspect	Lab 2 Initial Dialog Map (Analysis Model)	Lab 3 Refined Dialog Map (Design Model)	Design Rationale (Fox / Bruegge)
State Granularity	7 high-level states (S0–S6) with informal textual sub-variants.	**14 discrete design states**: 7 primary views, 4 modal overlays, and 3 transient/busy states.	Models modal lifecycles, dirty confirmation, and async gateway waits.
Transition Formalism	Informal natural language descriptions.	Strict UML syntax: Event [Guard] / Action -> Next State across 35 transitions.	Eliminates ambiguity for frontend implementation and Playwright E2E tests.
Component Binding	Abstract boundary references (D-00 to D-06).	1:1 binding to React components (PlugPlanApp, AuthScreen, JourneyMap).	Direct implementation traceability to TypeScript skeleton.
Modal Semantics	Treated as generic self-loops or textual side notes.	Formal modal overlays (ReportIssueModal, DuplicateAlertModal, ReportReviewModal).	Adheres to Fox p. 420 acceptor principles; retains parent view state.
State Persistence	Generic note regarding transient drafts.	Partitioned in-memory React state (Draft, dirty flag) vs. persistent PostgreSQL entities.	Prevents accidental navigation loss while preserving stateless backend APIs.
Concurrency & Resilience	Generic exception bullets.	Optimistic concurrency (HTTP 409 conflict), duplicate report alert, 3.9s gateway timeout, and cached fallback.	Resolves distributed data hazards per Bruegge Chapter 7.4.
Responsive Parity	Textual equivalence noted (M-01 = D-02).	Explicit mobile interaction models (accordion drawers, bottom nav bar, swipe cards).	WCAG 2.1 AA accessibility and touch target parity (>= 44x44px).

2. Global System Dialog Map — Role-Based Routing Architecture

The global system dialog map defines initial application bootstrap, credential verification in D-00 AuthScreen, and strict role-based workspace segregation. Driver sessions are routed to vehicle setup (D-01) or journey planning (D-02), while Moderator sessions are routed exclusively to the moderation dashboard (D-06). Global session invalidation (manual logout or HTTP 401 token expiry) revokes the Supabase JWT and resets client state to D-00.

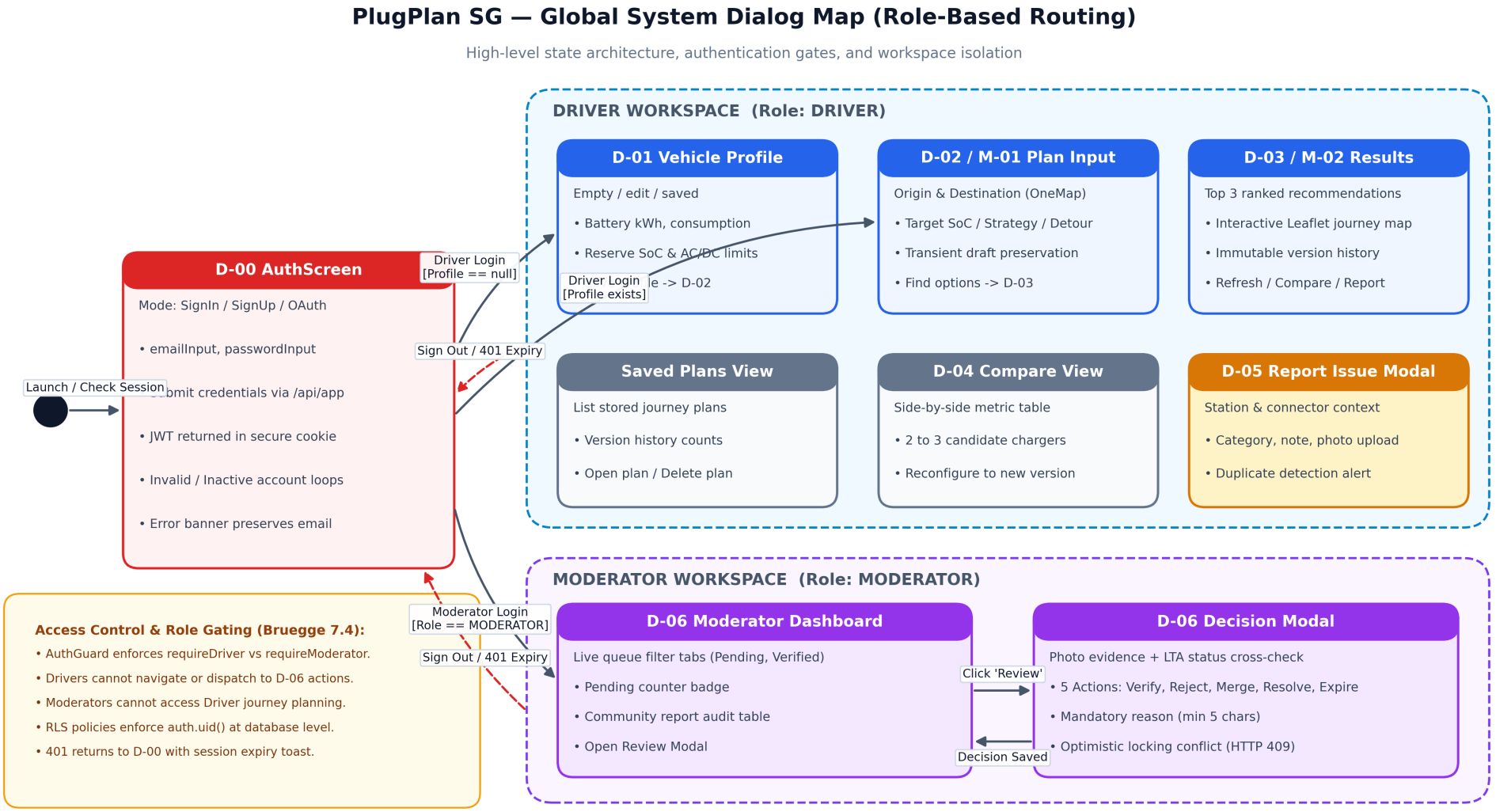


Figure 1: *Global System Dialog Map — Initial entry, authentication gates, workspace boundaries, and global session termination.*

3. Detailed Driver Journey Dialog Map (UC-02, UC-03, UC-04, UC-05, UC-06, UC-07)

The Driver journey models end-to-end charging decision support: profile management (D-01), journey planning (D-02), recommendation visualization via Leaflet (D-03), side-by-side multi-charger trade-off comparison (D-04), immutable version creation on reconfiguration (UC-05), manual data refresh with fault-tolerant fallback (UC-06), and community issue reporting (D-05). In-state transitions protect user drafts and provide clear validation feedback.

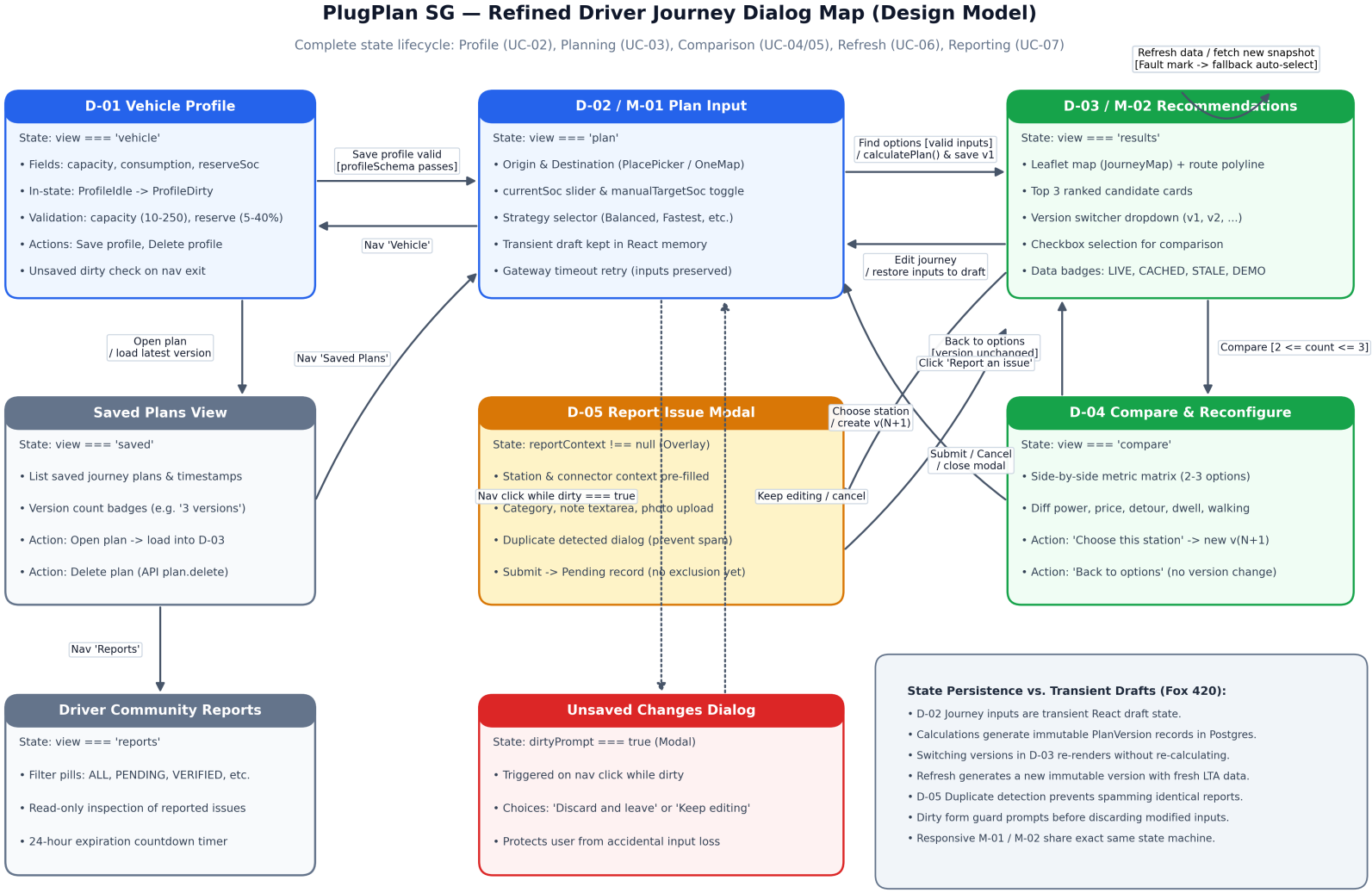


Figure 2: *Refined Driver Journey Dialog Map — Discrete design states, modal overlays, dirty-form guards, and versioned engine recalculations.*

4. Reporting & Moderator Journey Dialog Map (UC-07 & UC-08)

This state machine specifies community issue reporting, moderator queue auditing, and official status cross-referencing. A driver submission in D-05 stores a Pending record in Postgres without immediately excluding a station. When a Moderator adjudicates in D-06, mandatory reason validation (>= 5 chars) is enforced, and optimistic concurrency control (HTTP 409 conflict detection) prevents simultaneous overwrite hazards. A verified fault dynamically activates station exclusion in future calculations.

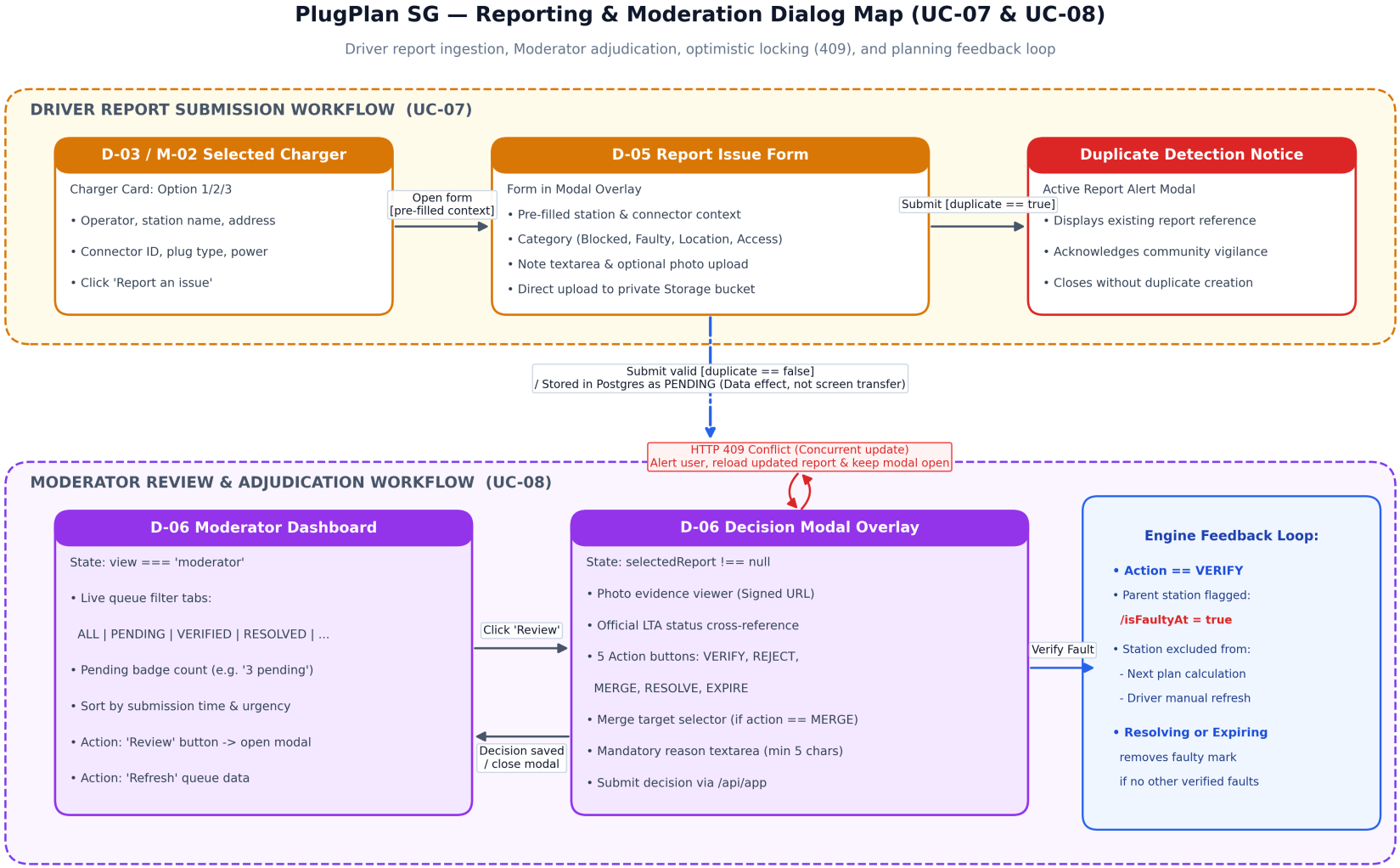


Figure 3: *Reporting and Moderation Dialog Map — Submission workflow, photo evidence review, optimistic locking loop, and planning engine feedback.*

5. Responsive & Mobile UI Dialog Adaptations (M-01 & M-02)

To satisfy NTU SC2006 requirements for multi-device support, PlugPlan SG provides responsive adaptations for handheld viewports (< 768px). While desktop frames (D-02, D-03, D-04) utilize split-screen layouts and permanent sidebars, mobile companions (M-01, M-02) utilize vertical accordion drawers, bottom navigation bars, and swipeable card stacks. Both viewports execute identical Zod validation schemas, business rules, and PlanEngine algorithms.

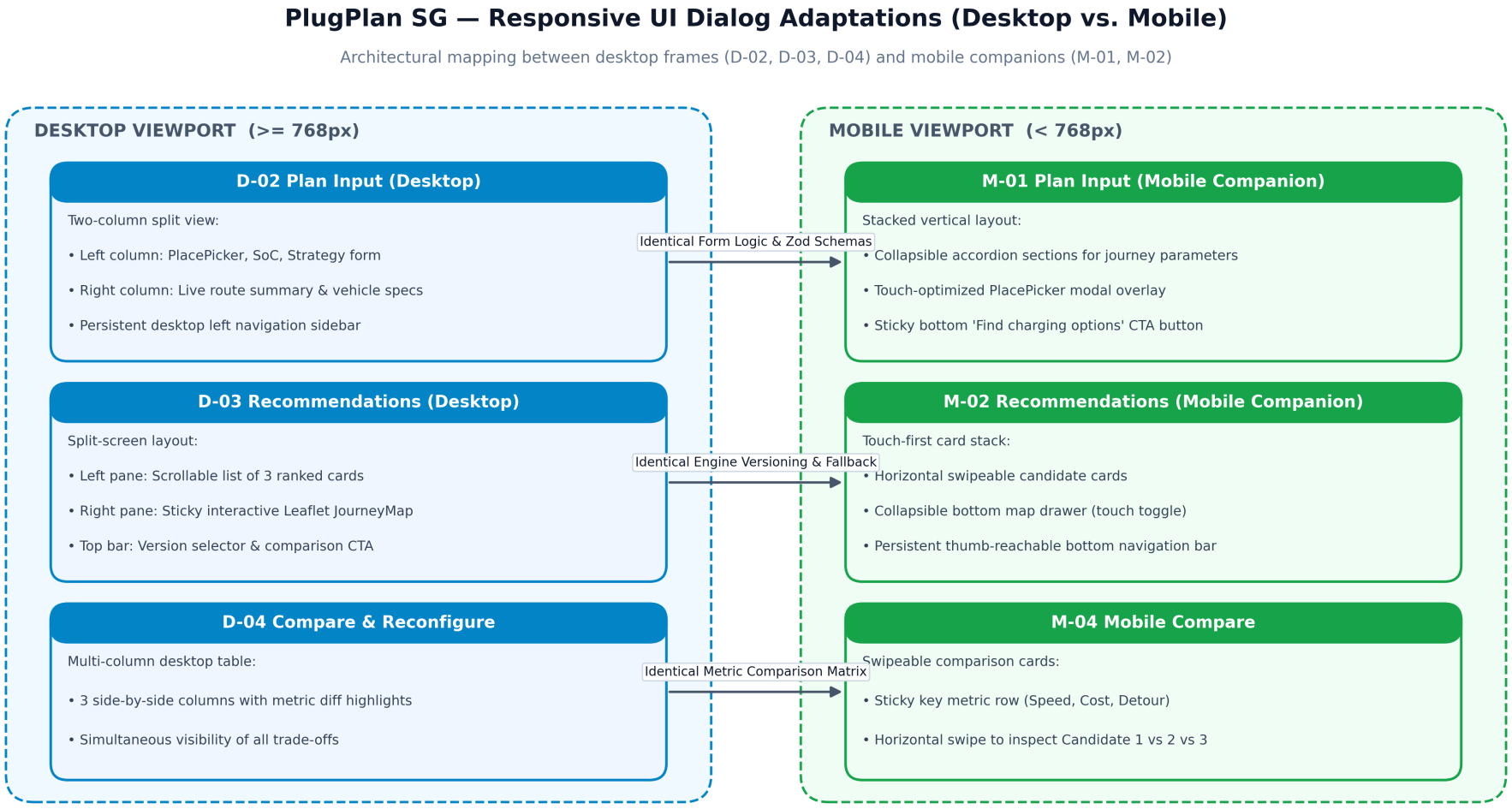


Figure 4: *Responsive UI Dialog Adaptations — Desktop split-screen vs. mobile touch-first card stacks and navigation parity.*

6. Comprehensive State Transition Specification Matrix

The following matrix formally specifies all 35 dialog transitions (T-01 through T-35). Each entry details trigger events, guard conditions, system actions / side effects, target states, bound design classes, and traceability.

ID	From	Trigger Event	Guard Condition	Action / Side Effect	To	Design Classes Involved	Trace
T-01	D-00 AuthScreen	Click 'Sign In'	Fields valid email format	ApiClient.api('auth.signin'); set session JWT	D-01 / D-02 / D-06	AuthScreen, ApiClient, AuthGuard	UC-01, FR-01
T-02	D-00 AuthScreen	Click 'Create Account'	Valid email, pwd >= 6	ApiClient.api('auth.signup'); init profile	D-01 Vehicle Profile	AuthScreen, ApiClient, AuthGuard	UC-01, FR-02
T-03	D-00 AuthScreen	Submit invalid credentials	Auth error (400/401)	Display inline error banner; preserve email	D-00 SignInError	AuthScreen, ErrorMapper	UC-01 (Ex)
T-04	Any Auth State	Click 'Sign out'	User confirmed	SupabaseClient.auth.signOut(); clear JWT & local state	D-00 AuthScreen	PlugPlanApp, SupabaseClient	UC-01
T-05	Any Auth State	API returns HTTP 401	JWT expired / revoked	Clear session; toast 'Session expired, please sign in'	D-00 AuthScreen	ApiClient, AuthGuard, PlugPlanApp	UC-01
T-06	D-01 Vehicle Profile	Modify input field	Value changed	Set dirty = true; update profileDraft in React state	D-01 ProfileDirty	PlugPlanApp	UC-02
T-07	D-01 Vehicle Profile	Click 'Save vehicle'	profileSchema passes	ApiClient.api('profile.save'); set dirty=false; toast	D-02 Plan Input	PlugPlanApp, PlanRepository	UC-02, FR-03/04
T-08	D-01 Vehicle Profile	Click 'Save vehicle'	Validation fails	Highlight field errors (capacity 10-250, reserve 5-40%)	D-01 ProfileValidationError	PlugPlanApp, ValidationSchemas	UC-02 (Ex)
T-09	D-01 Vehicle Profile	Click 'Delete profile'	Confirmed by user	ApiClient.api('profile.delete'); reset draft to defaults	D-01 ProfileIdle (Empty)	PlugPlanApp, PlanRepository	UC-02
T-10	D-02 Plan Input	Type place query	Query >= 2 chars	Debounced search via ProviderGateway.searchPlaces()	D-02 AutocompleteActive	PlacePicker, ProviderGateway	UC-03, FR-05
T-11	D-02 Plan Input	Select place suggestion	Item clicked	Set origin/destination coords & postal; close dropdown	D-02 PlaceSelected	PlacePicker, PlugPlanApp	UC-03, FR-05
T-12	D-02 Plan Input	Click 'Find options'	planInputSchema passes	Set busy=true; ApiClient.api('plan.create'); save v1	D-03 Recommendations	PlugPlanApp, PlanEngine	UC-03, FR-05-19
T-13	D-02 Plan Input	Click 'Find options'	Form invalid	Display field validation errors; focus first invalid field	D-02 InputValidationError	PlugPlanApp, ValidationSchemas	UC-03 (Ex)
T-14	D-02 Plan Input	Click 'Find options'	Gateway timeout (3.9s)	Catch ApiError (503/504); display error banner; retain inputs	D-02 ProviderErrorState	PlugPlanApp, ProviderGateway	UC-03 (Ex)
T-15	D-02 Plan Input	Nav click while dirty	dirty === true	Open UnsavedChangesPromptModal ('Discard inputs?')	UnsavedChangesModal	PlugPlanApp	UC-03
T-16	D-03 Results	Click 'Edit journey'	Plan version loaded	Set draft = version.inputs; set editPlanId; navigate	D-02 Plan Input	PlugPlanApp	UC-05
T-17	D-03 Results	Toggle charger checkbox	Card clicked	If selected: remove; if unselected & count < 3: add	D-03 CardSelection	PlugPlanApp	UC-04, FR-20
T-18	D-03 Results	Click 'Compare options'	2 <= count <= 3	Filter recommendations by compareIds; switch view	D-04 CompareView	PlugPlanApp	UC-04, FR-20
T-19	D-03 Results	Change version dropdown	Version selected	Set activeVersionId = id; re-render JourneyMap & cards	D-03 DisplayShortlist	PlugPlanApp, JourneyMap	UC-05, FR-21
T-20	D-03 Results	Click 'Refresh data'	Network active	Set busy=true; ApiClient.api('plan.refresh'); new snapshot	D-03 (New Version)	PlugPlanApp, PlanEngine	UC-06, FR-22/23
T-21	D-03 Results	Refresh with fault marked	Station verified faulty	PlanEngine excludes station; auto-selects fallback	D-03 RefreshFallback	PlanEngine, ModeratorFilter	UC-06, FR-24
T-22	D-03 Results	Click 'Report an issue'	Card clicked	Set reportContext = { station, connector }; open modal	D-05 ReportIssueModal	PlugPlanApp, Modal	UC-07
T-23	D-04 CompareView	Click 'Back to options'	Comparison active	Switch view = 'results'; active version preserved	D-03 Recommendations	PlugPlanApp	UC-04
T-24	D-04 CompareView	Click 'Choose station'	Candidate selected	Set recommendation primary; generate new PlanVersion v(N+1)	D-03 (New Version)	PlugPlanApp, PlanRepository	UC-05, FR-21
T-25	D-05 Report Modal	Select photo file	File <= 5MB, image/*	Read file; generate preview URL; store File in state	D-05 FileSelected	PlugPlanApp, SupabaseClient	UC-07
T-26	D-05 Report Modal	Click 'Submit report'	Category valid, note ok	Upload photo; ApiClient.api('report.submit'); status PENDING	D-05 ReportSuccessToast	PlugPlanApp, ReportRepository	UC-07, FR-24
T-27	D-05 Report Modal	Click 'Submit report'	Active report exists	API returns { duplicate: true }; open duplicate notice	D-05 DuplicateNotice	PlugPlanApp, Modal	UC-07 (Ex)

ID	From	Trigger Event	Guard Condition	Action / Side Effect	To	Design Classes Involved	Trace
T-28	D-05 Report Modal	Click 'Cancel' / Backdrop	No upload in progress	Reset reportContext = null; close modal without saving	D-03 Recommendations	PlugPlanApp	UC-07
T-29	D-06 Moderator	Click status filter tab	Tab clicked	Filter local reports array by statusLabel(r) === tab	D-06 QueueFilterTabs	PlugPlanApp	UC-08
T-30	D-06 Moderator	Click 'Review' on report	Moderator auth ok	Set selectedReport = r; init action 'VERIFY'; open modal	D-06 ReviewModal	PlugPlanApp, Modal	UC-08
T-31	D-06 ReviewModal	Click 'Confirm decision'	reason.length >= 5	ApiClient.api('report.moderate'); record decision; toast	D-06 DecisionSuccess	PlugPlanApp, ReportRepository	UC-08, FR-25
T-32	D-06 ReviewModal	Click 'Confirm decision'	reason.length < 5	Display inline error: 'Justification reason is required'	D-06 ReasonError	PlugPlanApp	UC-08 (Ex)
T-33	D-06 ReviewModal	Click 'Confirm decision'	Concurrency conflict	Catch HTTP 409 Conflict; toast alert; reload report data	D-06 ConcurrencyConflict	PlugPlanApp, ErrorMapper	UC-08 (Ex)
T-34	Saved Plans View	Click 'Open plan'	Plan card clicked	Set currentPlanId; set activeVersionId; view='results'	D-03 Recommendations	PlugPlanApp, JourneyMap	UC-03, UC-04
T-35	Saved Plans View	Click 'Delete plan'	Deletion confirmed	ApiClient.api('plan.delete'); remove from list; toast alert	Saved Plans View	PlugPlanApp, PlanRepository	UC-03

7. UI State Specifications & Christopher Fox Callouts (Fox p. 420)

Christopher Fox (2006, p. 420) mandates UI callouts to document visual elements, component hierarchies, input constraints, and dynamic enabling/disabling logic that cannot be expressed through state nodes alone.

Callout C-01: Global Workspace Shell & Navigation

- Permanent Left Sidebar: Houses workspace navigation (Plan, Vehicle, Saved Plans, Community Reports, Moderation Queue).
- Session Chip & Role Badge: Displays user email and authorization role badge ('DRIVER' | 'MODERATOR').
- Dynamic Counters: Saved Plans displays count badge of active plans; Moderation Queue displays count badge of PENDING reports.
- Sign Out Button: Revokes Supabase session JWT, invalidates client state, and navigates immediately to D-00.

Callout C-02: D-00 Account Access (AuthScreen)

- Mode Toggle: Instant client switch between 'Sign In' and 'Create Account' modes without page reloads.
- Form Controls: Email input (HTML5 type='email', required), password input (type='password', minlength='6', required).
- Disabled State: 'Sign In' button is disabled while an API request is in flight (busy === true).
- Security: Role-agnostic error messages prevent username enumeration on authentication failures.

Callout C-03: D-01 Vehicle Profile View (VehicleProfileView)

- Usable Battery Capacity: Numeric input constrained to 10.0–250.0 kWh (step 0.1).
- Energy Consumption: Numeric input constrained to 5.0–50.0 kWh per 100 km.
- Reserve State of Charge: Slider + numeric input constrained to 5%–40% (default 15%).
- Detour Tolerance: Numeric input constrained to 0–45 minutes (default 15 min).
- Charging Power Limits: AC power (3.0–43.0 kW) and DC power (0.0–350.0 kW).
- Plug Types: Multi-toggle selector (Type 2, CCS2, CHAdeMO); at least one plug type must be active.
- Dirty State Guard: Unsaved changes prompt modal is displayed if user navigates away while dirty === true.

Callout C-04: D-02 / M-01 Plan Input View (PlanInputView)

- Origin & Destination: PlacePicker component with debounced OneMap Geocoding API integration (lat, lng, postal).
- Target SoC Toggle: Smart Arrival Reserve (distance-based destination target) vs. Manual Target % (20%–100%).
- Strategy Radio Cards: Balanced (default), Fastest (DC prioritized), Cheapest ($/kWh prioritized), Max Availability, Min Detour.
- Disabled State: 'Find charging options' button is disabled until both coordinates are valid and vehicle profile is loaded.

Callout C-05: D-03 / M-02 Charging Recommendations View (ResultsView)

- Data State Badges: LIVE (green), CACHED (blue), STALE (amber with provider warning), DEMO_FIXTURE (purple).
- Version History Selector: Dropdown of all immutable versions for this plan (e.g. 'Version 2 · refreshed').
- Interactive Leaflet JourneyMap: Green start pin, red finish pin, decoded route polyline, numbered charger candidate markers.
- Recommendation Cards: Rank, Station Name, Address, Connector kW, Availability, Detour Km/Min, Dwell Min, Cost ($ SGD).
- Comparison Checkboxes: Allows selecting up to 3 candidate cards. 'Compare' button enabled if and only if 2 <= count <= 3.

Callout C-06: D-04 Compare & Reconfigure View (CompareView)

- Multi-Column Matrix: Side-by-side comparison of 2 or 3 selected recommendations across identical metrics.
- Dwell vs. Detour Diff: Color-coded delta highlights comparing detour penalty against charging speed.
- Action 'Choose this station': Reconfigures the journey around the chosen stop, creating an immutable PlanVersion v(N+1).
- Action 'Back to options': Returns to D-03 without creating a new version.

Callout C-07: D-05 Report Issue Modal (ReportIssueModal)

- Pre-filled Context: Station name, address, operator, and connector ID populated automatically from parent recommendation card.
- Category Dropdown: BLOCKED_BAY, SUSPECTED_FAULTY_CONNECTOR, INCORRECT_LOCATION, ACCESS_PROBLEM.
- Free-text Note: Description textarea (max 500 characters).
- Photo Evidence: File upload with client image preview (JPEG/PNG <= 5MB); uploaded to private Supabase Storage bucket.
- Duplicate Alert: Polite dialog acknowledging community vigilance if an active report already exists within 24 hours.

Callout C-08: D-06 Moderator Dashboard & Decision Modal (ModeratorReviewView)

- Queue Filter Tabs: ALL, PENDING, VERIFIED, RESOLVED, REJECTED, MERGED, EXPIRED with dynamic pending badge counter.
- Evidence Inspection: High-resolution photo evidence rendered via secure time-limited Supabase Storage signed URLs.
- Official Status Cross-Reference: Displays live LTA DataMall charger operational status alongside community complaint.
- Decision Actions: Five explicit buttons (VERIFY, REJECT, MERGE, RESOLVE, EXPIRE).
- Mandatory Justification: Reason textarea must contain at least 5 characters; submission button is disabled otherwise.
- Concurrency Alert: Intercepts HTTP 409 conflict if another moderator acted simultaneously; prompts refresh.

8. Key Design Issues Resolution (Bruegge Chapter 7.4)

8.1 Access Control and Security

Authentication is delegated to Supabase Auth using JSON Web Tokens (JWT) stored in HTTP-only secure session cookies. The application enforces defense-in-depth: the presentation layer gates navigation between Driver views and Moderator views via AuthGuard, the application router checks requireDriver and requireModerator on incoming requests, and the PostgreSQL persistence layer enforces Row-Level Security (RLS) policies matching auth.uid() on all database operations.

8.2 Persistent vs. Transient State Management

A rigorous boundary is maintained between transient frontend drafts and persistent backend entities. User inputs in D-02 (origin, destination, target SoC) reside in React in-memory state (Draft). If a route search fails or place ambiguity arises, draft inputs are preserved in memory to prevent input fatigue. Once calculated, results are persisted as immutable PlanVersion entities in PostgreSQL. Navigating away from modified profiles or inputs triggers the UnsavedChangesModal.

8.3 Concurrency Control (Optimistic Locking & HTTP 409)

In the community moderation workflow (UC-08), multiple moderators may review the queue concurrently. The database RPC moderate_report checks the report's current status and revision timestamp before committing decisions. If another moderator has updated the report, the RPC rejects the transaction with HTTP 409 Conflict. The dialog map catches this exception, keeps the modal open, displays an alert, and reloads the fresh report record.

8.4 Degraded Mode & Provider Resilience

To safeguard against external API outages (OneMap Geocoder/Routing or LTA DataMall), ProviderGateway enforces a strict 3.9-second deadline budget. If live calls timeout or fail, the system falls back to the disk cache (.cache/lta-public.json) and tags the snapshot as STALE, allowing drivers to complete planning while displaying an explicit amber warning badge in D-03.

9. Comprehensive Requirements & Class Traceability Matrix

Use Case	UI Frames	Design Classes Involved	Architectural Layer	Functional Requirements
UC-01 Access Account	D-00	AuthScreen, ApiClient, RouteHandler, AuthGuard, SupabaseClient	Presentation, App, Persistence	FR-01, FR-02
UC-02 Manage EV Profile	D-01	PlugPlanApp, ApiClient, PlanRepository, ValidationSchemas	Presentation, App, Domain, Persistence	FR-03, FR-04
UC-03 Create Charging Plan	D-02, D-03, M-01, M-02	PlugPlanApp, PlacePicker, JourneyMap, PlanEngine, ProviderGateway	Presentation, App, Domain, Integration	FR-05 through FR-19
UC-04 Review & Compare	D-03, D-04, M-02	PlugPlanApp, JourneyMap, ApiClient	Presentation Layer	FR-20
UC-05 Reconfigure Plan	D-04, D-03	PlugPlanApp, PlanEngine, PlanRepository	Presentation, Domain, Persistence	FR-21
UC-06 Refresh & Fallback	D-03, M-02	PlugPlanApp, PlanEngine, ModeratorFaultFilter, PlanRepository	Presentation, Domain, Integration	FR-22, FR-23, FR-24
UC-07 Submit Charger Issue	D-05	PlugPlanApp, Modal, ApiClient, ReportRepository, SupabaseClient	Presentation, App, Persistence	FR-24
UC-08 Moderate Reports	D-06	PlugPlanApp, Modal, ReportRepository, ModerationControl, ModerationDecision	Presentation, App, Domain, Persistence	FR-25